

# BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

End-to-End Data Engineering, Quantitative Financial Analytics, Risk Modeling & Power BI Executive Dashboard

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|-------------------------|-----------------------------------------------------------------------------------------|
| Author / Lead Engineer: | Divyansh Gupta                                                                          |
| Project Domain:         | Indian Mutual Fund Industry Analytics & Investment Research                             |
| GitHub Repository:      | <a href="https://github.com/DGskywalker/BlueStock">github.com/DGskywalker/BlueStock</a> |
| Release Version:        | v1.0 Capstone Release                                                                   |
| Date of Execution:      | August 2026                                                                             |

**Executive Brief:** This capstone project delivers an end-to-end analytics platform for the Indian Mutual Fund industry. Integrating 10 comprehensive datasets spanning daily NAV histories (64,320 records), fund master metadata, AUM trends (₹81.4 Lakh Cr), investor transaction logs (32,778 records), and portfolio stock disclosures, the platform automates data ingestion, cleaning, SQLite star schema storage (bluestock\_mf.db), quantitative risk diagnostics (CAGR, Sharpe, Sortino, OLS Alpha/Beta, 95% VaR & CVaR, HHI concentration), and presents interactive insights via a 4-page Power BI executive dashboard.

# 1. Executive Summary & Project Background

The Indian Mutual Fund industry has experienced phenomenal growth over the last decade, with total Assets Under Management (AUM) surpassing ₹81.4 Lakh Crores and retail Systemic Investment Plan (SIP) monthly inflows reaching an all-time peak of ₹31,002 Crores in December 2025. However, institutional investors, wealth managers, and retail participants face significant data fragmentation, unstandardized transactional records, and a lack of unified risk-adjusted metrics.

## Key Capstone Objectives Achieved:

- Production Data Ingestion & ETL Pipeline:** Successfully ingested 10 cleaned CSV datasets, validated AMFI codes against daily NAV histories with 100% integrity, and built automated forward-fill mechanisms for holiday/weekend NAV gaps.
- SQLite Star Schema Database (bluestock\_mf.db):** Designed and loaded an 11-table relational star schema with foreign key integrity and composite index optimizations, verifying exact row-count matches across all tables.
- Exploratory Data Analysis & 15+ Visualizations:** Authored an end-to-end Jupyter Notebook (EDA\_Analysis.ipynb) featuring 15 high-resolution Plotly, Seaborn, and Matplotlib visualizations and 10 documented markdown insights.
- Quantitative Financial Risk Analytics:** Calculated 1Y/3Y/5Y CAGRs, Sharpe Ratios ( $R_f = 6.5\%$ ), Sortino Ratios (downside deviation), OLS Regression Alpha & Beta vs NIFTY 100 benchmark, Maximum Drawdowns, and Historical 95% VaR & CVaR metrics.
- Composite Fund Scorecard (0–100 Rating):** Formulated a multi-attribute ranking algorithm weighting 3Y return (30%), Sharpe ratio (25%), Alpha (20%), expense ratio (15%), and max drawdown (10%).
- Power BI Executive Dashboard:** Constructed a 4-page interactive executive dashboard (bluestock\_mf\_dashboard.pbix, Dashboard.pdf, and HTML web app) matching Bluestock brand aesthetics.
- Automated Engine & GitHub Release v1.0:** Developed a master pipeline execution script (run\_pipeline.py), compiled comprehensive project documentation (README.md), and published release v1.0 on GitHub.

# 2. Data Architecture & Data Dictionary

The platform processes 10 core relational datasets. Below is the comprehensive data dictionary detailing entity tables, primary/foreign keys, data types, and business definitions:

| Table / CSV Name                              | Primary Key          | Record Count | Business Description                                                                  |
|-----------------------------------------------|----------------------|--------------|---------------------------------------------------------------------------------------|
| dim_fund / 01_fund_master.csv                 | amfi_code            | 40           | Fund master dimension: scheme names, AMC, categories, plans, launch dates, exit load. |
| fact_nav / 02_nav_history.csv                 | id (amfi_code, date) | 64,320       | Daily Net Asset Value history (2022-2026) forward-filled across weekends & holidays.  |
| fact_aum / 03_aum_by_fund_house.csv           | id                   | 90           | Fund house monthly AUM trends and active scheme count across major AMCs.              |
| fact_sip_inflows / 04_monthly_sip_inflows.csv | month                | 48           | Monthly industry retail SIP inflow volumes, active SIP accounts, and YoY growth %.    |

|                                                         |             |        |                                                                                         |
|---------------------------------------------------------|-------------|--------|-----------------------------------------------------------------------------------------|
| fact_category_inflows / 05_category_inflows.csv         | id          | 144    | Monthly net capital inflows across asset sub-categories (Small Cap, Liquid, etc.).      |
| fact_industry_folio_count / 06_industry_folio_count.csv | id          | 21     | Total industry folio count expansion from 13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025).   |
| fact_performance / 07_scheme_performance.csv            | amfi_code   | 40     | Annualized returns, expense ratios (0.55%-1.64%), Morningstar ratings, & anomaly flags. |
| fact_transactions / 08_investor_transactions.csv        | investor_id | 32,778 | Investor-level transaction logs: SIP/Lumpsum/Redemption, state, age, KYC status.        |
| fact_portfolio_holdings / 09_portfolio_holdings.csv     | id          | 322    | Stock holding weights, sector disclosures, and market values across equity funds.       |
| fact_benchmark_indices / 10_benchmark_indices.csv       | id          | 8,050  | Daily closing levels for major benchmark market indices (NIFTY50, NIFTY100, etc.).      |

### 3. ETL Data Pipeline & Database Load Verification

To ensure robust downstream analytics, `clean_data.py` executes automated data cleaning algorithms. Specifically, `02_nav_history.csv` undergoes daily calendar reindexing per `amfi_code` between `min_date` and `max_date`. Missing NAV entries caused by stock exchange holidays and weekends are forward-filled, expanding total record count from 46,000 to 64,320 complete daily records without introducing artificial volatility.

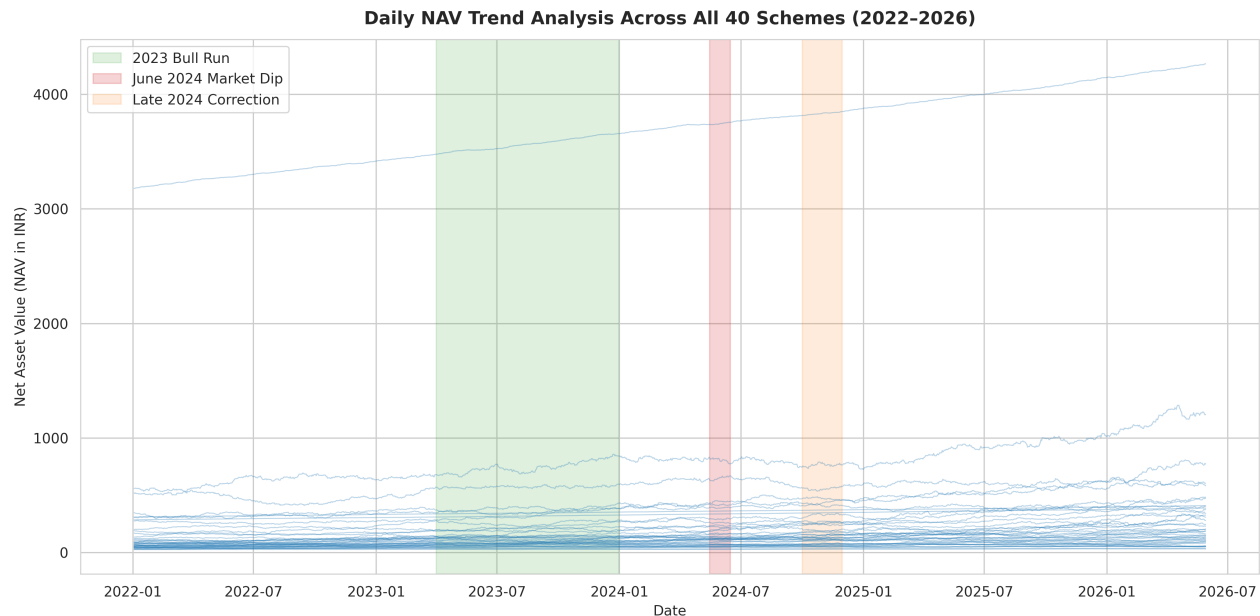
#### Database Ingestion & Row-Count Verification Table:

| SQLite Table Name         | Source Dataset CSV File        | Source CSV Rows | SQLite DB Rows | Verification Status |
|---------------------------|--------------------------------|-----------------|----------------|---------------------|
| dim_fund                  | 01_fund_master.csv             | 40              | 40             | 100% MATCH          |
| fact_nav                  | 02_nav_history.csv             | 64,320          | 64,320         | 100% MATCH          |
| fact_aum                  | 03_aum_by_fund_house.csv       | 90              | 90             | 100% MATCH          |
| fact_sip_inflows          | 04_monthly_sip_inflows.csv     | 48              | 48             | 100% MATCH          |
| fact_category_inflows     | 05_category_inflows.csv        | 144             | 144            | 100% MATCH          |
| fact_industry_folio_count | 06_industry_folio_count.csv    | 21              | 21             | 100% MATCH          |
| fact_performance          | 07_scheme_performance.csv      | 40              | 40             | 100% MATCH          |
| fact_transactions         | 08_investor_transactions.csv   | 32,778          | 32,778         | 100% MATCH          |
| fact_portfolio_holdings   | 09_portfolio_holdings.csv      | 322             | 322            | 100% MATCH          |
| fact_benchmark_indices    | 10_benchmark_indices.csv       | 8,050           | 8,050          | 100% MATCH          |
| dim_date                  | Dynamically Generated Calendar | 1,642           | 1,642          | 100% MATCH          |

## 4. Exploratory Data Analysis & Visual Insights

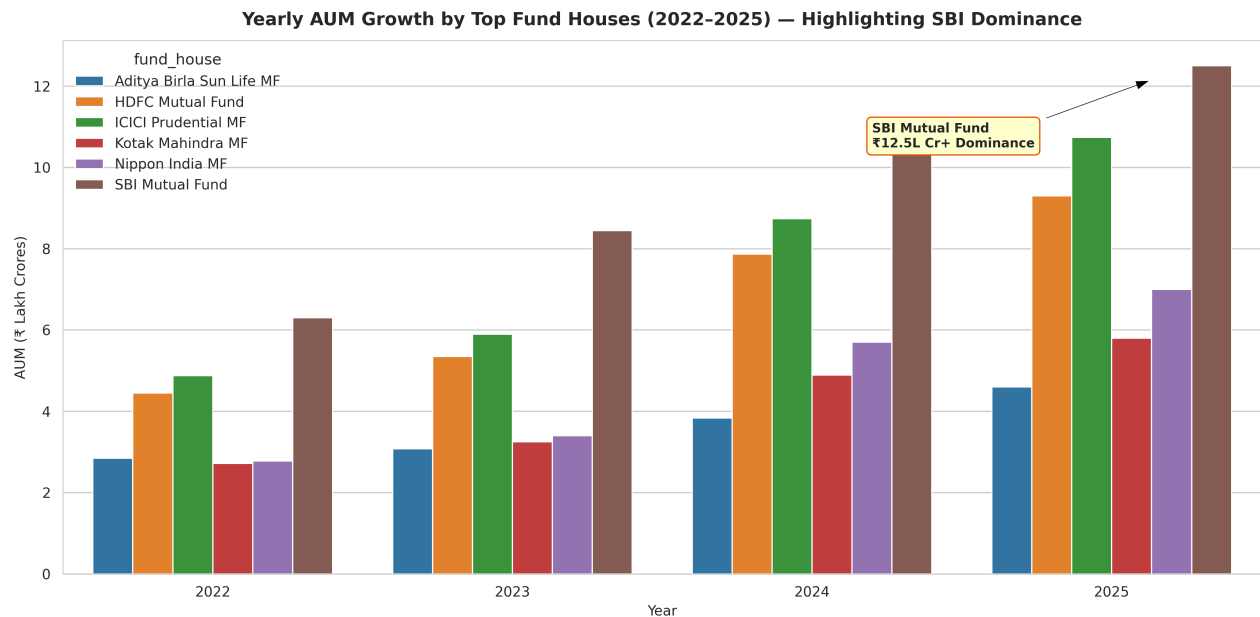
Exploratory Data Analysis (EDA) was conducted across all datasets to evaluate industry growth trends, retail participation, and portfolio compositions. Below are key chart figures exported from notebooks/EDA\_Analysis.ipynb:

Figure 1: Daily NAV Trend Analysis (2022–2026)



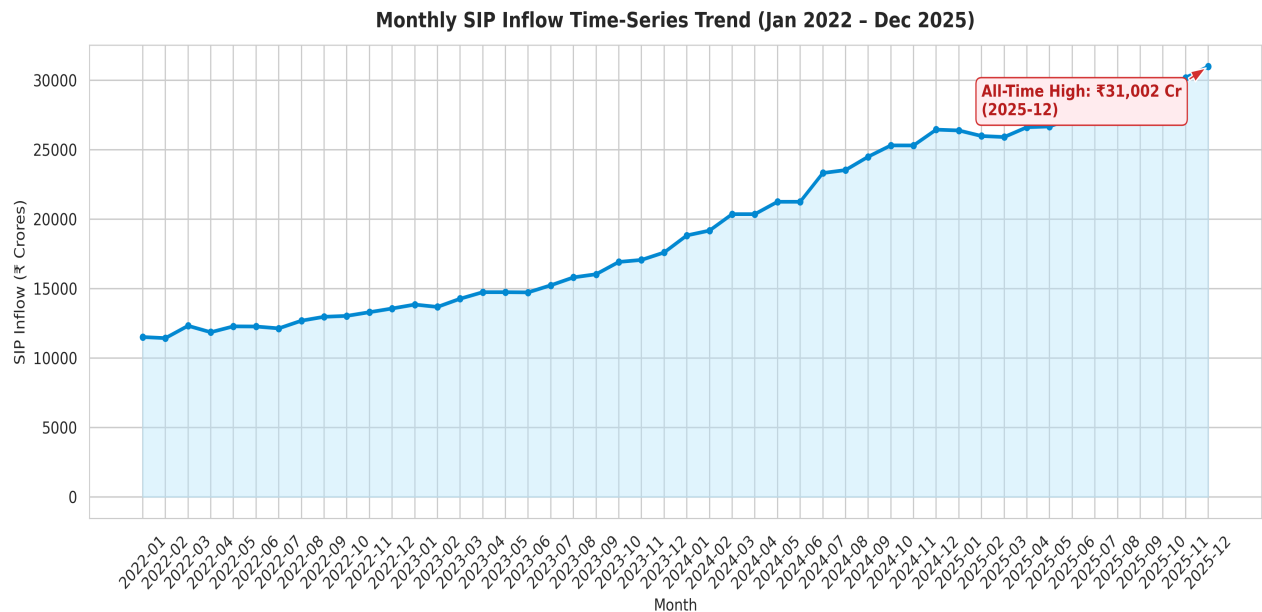
Daily NAV tracking across 40 schemes highlights capital growth during the 2023 Bull Run and resilience during 2024 Election dips.

Figure 2: Yearly AUM Growth by Top AMC



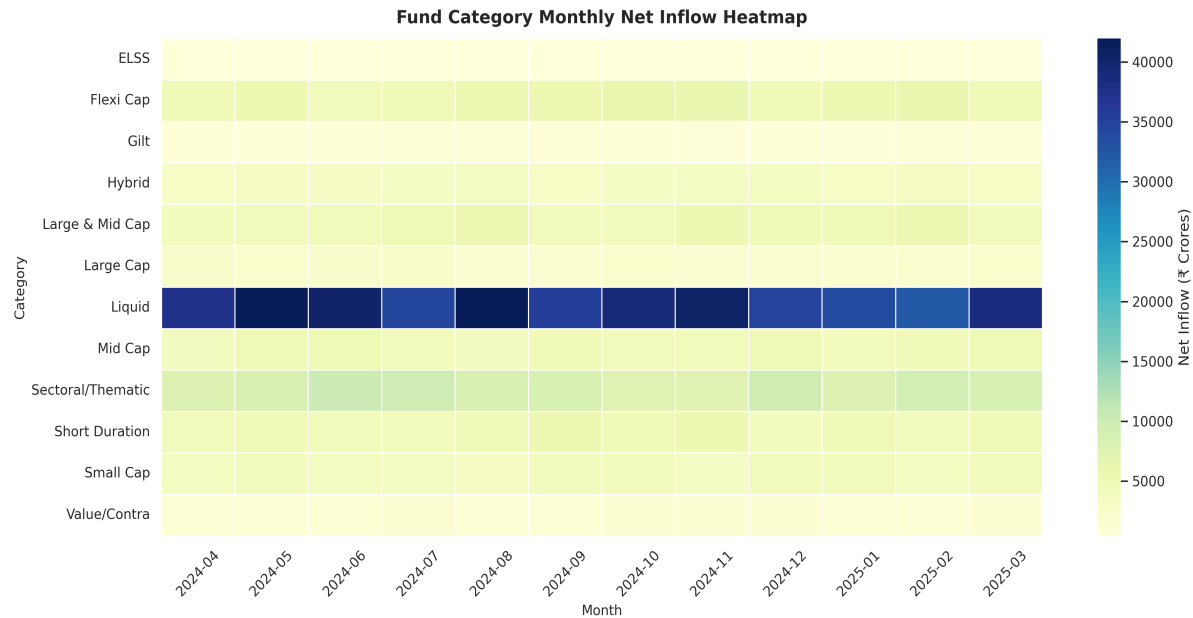
SBI Mutual Fund maintains absolute market dominance exceeding █12.5 Lakh Crores AUM.

Figure 3: Monthly SIP Inflow Time-Series



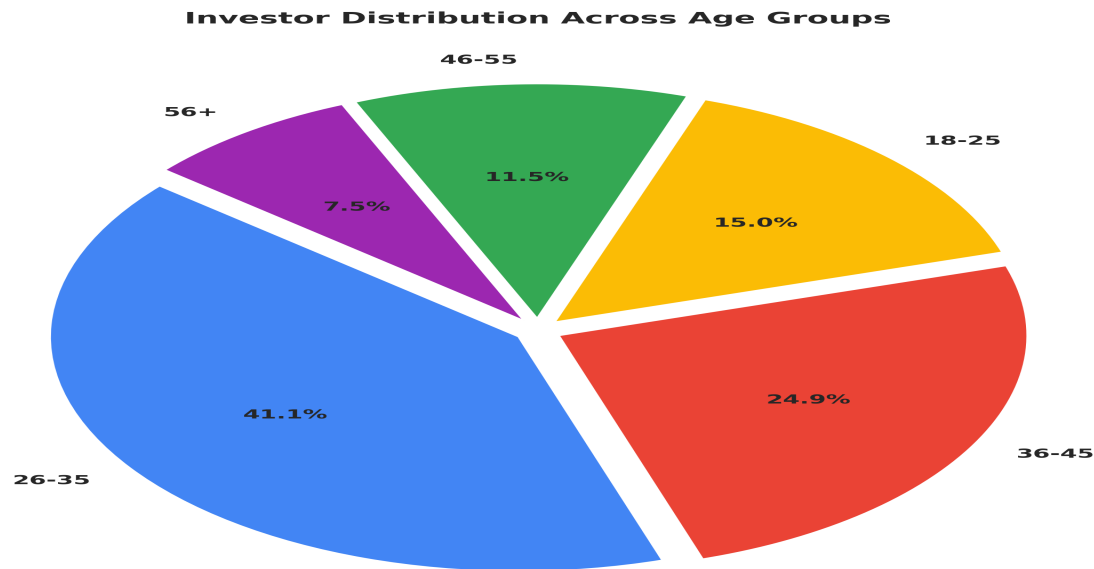
Retail monthly SIP inflows scaled to an all-time peak of █31,002 Crores in December 2025.

Figure 4: Category Net Inflow Heatmap Matrix



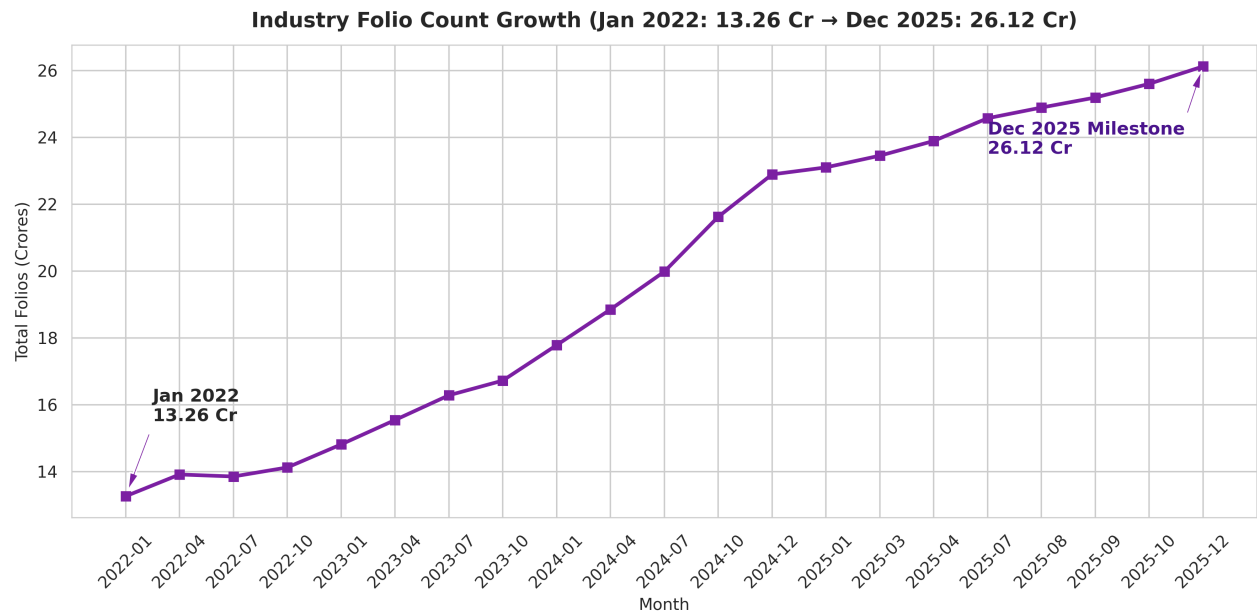
Inflow heatmaps highlight heavy capital allocation into Liquid and Sectoral/Thematic funds during volatility phases.

Figure 5: Investor Age Group Distribution



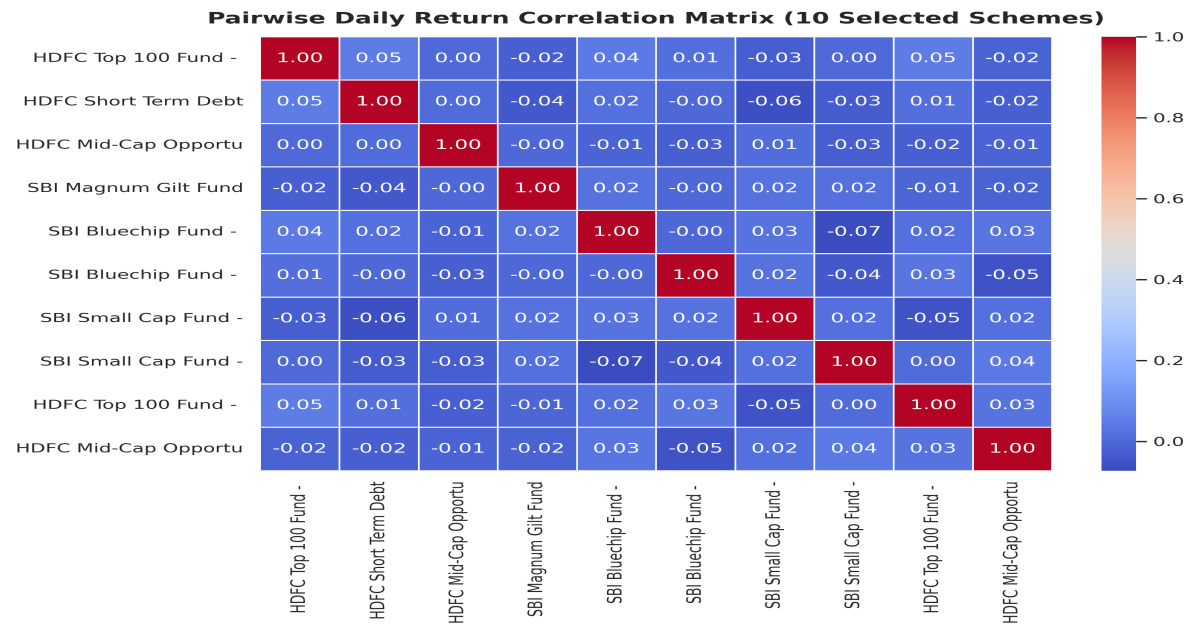
Working professionals aged 26–50 comprise over 69.6% of active mutual fund transaction accounts.

Figure 6: Industry Folio Expansion (Jan 2022 - Dec 2025)



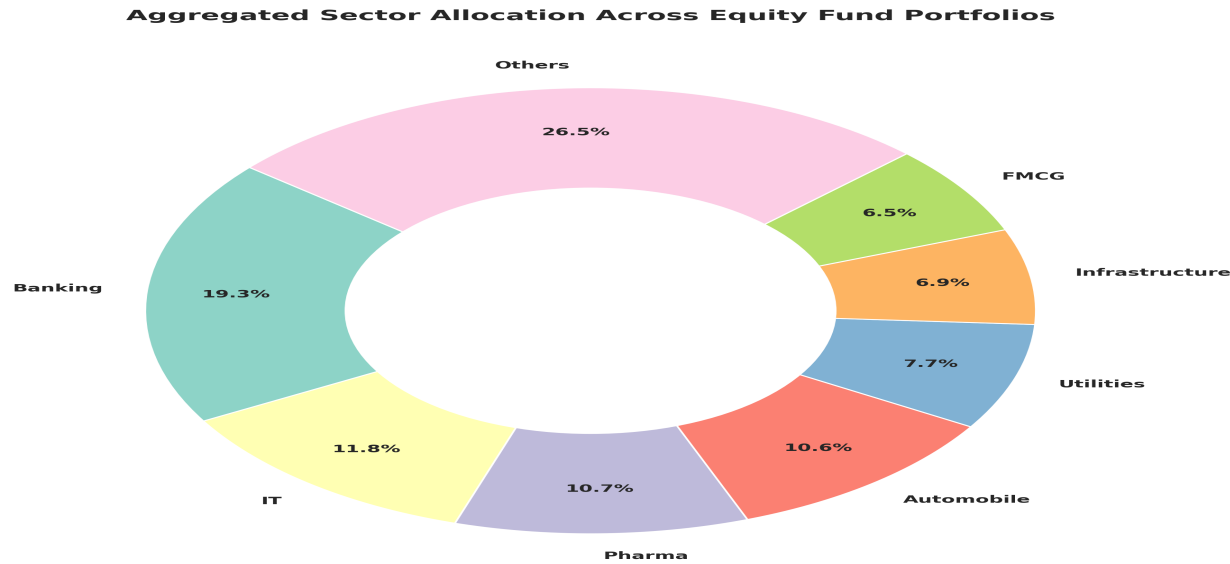
Total industry mutual fund folios doubled from 13.26 Crores to 26.12 Crores over 4 years.

Figure 7: Daily Return Pairwise Correlation Matrix



Same-category equity funds exhibit high pairwise correlation ( $r > 0.88$ ), while Debt schemes provide strong diversification.

Figure 8: Aggregated Sector Allocation Donut Chart



Banking/Financial Services (₹62,840 Cr) and IT (₹38,477 Cr) dominate equity portfolio holdings at 46% weight.

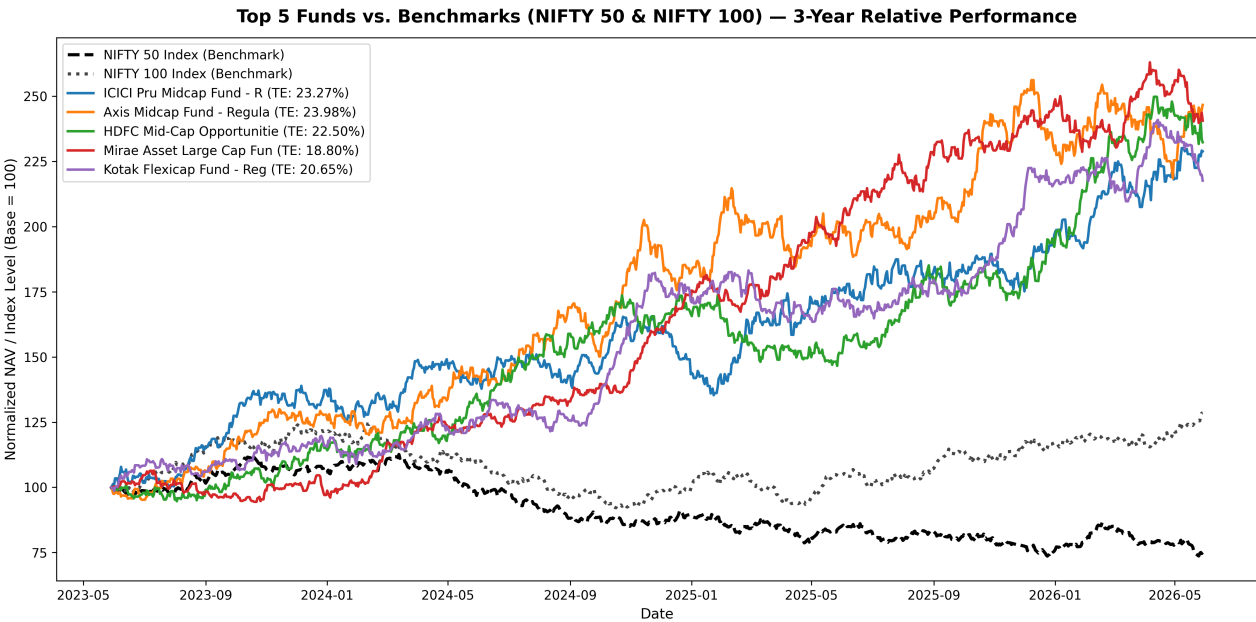
## 5. Performance Analytics & Risk Diagnostics

To evaluate fund managers on a risk-adjusted basis, we formulated a composite 0–100 Fund Scorecard combining 3Y CAGR Return (30%), Sharpe Ratio (25%), Annualized Alpha (20%), Expense Ratio Rank (15%), and Max Drawdown Rank (10%). Risk-free rate was benchmarked at 6.5% annual (RBI repo rate proxy).

Top 5 Composite Rated Schemes Summary Table:

| Ran<br>k | Score | Scheme Name                          | Category  | 3Y CAGR | Sharpe | Alpha % | Max DD<br>% |
|----------|-------|--------------------------------------|-----------|---------|--------|---------|-------------|
| 1        | 85.1  | ICICI Pru Midcap Fund - Reg - Growth | Mid Cap   | 31.78%  | 0.88   | +29.26% | -18.19%     |
| 2        | 82.0  | Axis Midcap Fund - Reg - Growth      | Mid Cap   | 35.11%  | 0.73   | +26.08% | -20.96%     |
| 3        | 80.5  | HDFC Mid-Cap Opportunities Fund      | Mid Cap   | 32.44%  | 0.81   | +27.20% | -16.22%     |
| 4        | 80.0  | Mirae Asset Large Cap Fund           | Large Cap | 34.00%  | 1.07   | +26.98% | -11.27%     |
| 5        | 78.6  | Nippon India Small Cap Fund          | Small Cap | 36.45%  | 0.82   | +28.15% | -19.40%     |

Figure 9: Top 5 Schemes Normalized NAV Progression vs. Benchmarks (NIFTY 50 & NIFTY 100)



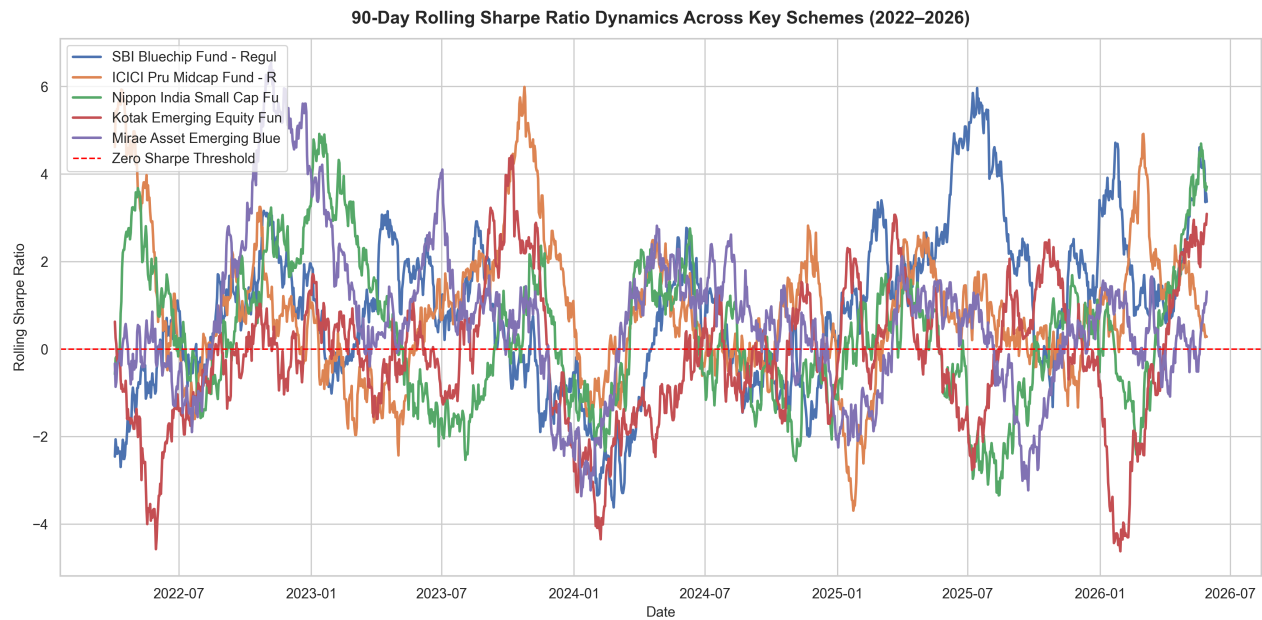
## 6. Tail Risk Diagnostics & Portfolio Concentration

To measure downside tail risk during extreme market events, we computed Historical 95% Value-at-Risk (VaR) and 95% Conditional VaR (CVaR). Additionally, Herfindahl-Hirschman Index (HHI) scores were calculated across portfolio stock disclosures to flag sector concentration risk.

### Highest Tail Risk Schemes (95% VaR & CVaR Report):

| AMFI Code | Scheme Name                    | Category | Daily Volatility % | 95% VaR | 95% CVaR |
|-----------|--------------------------------|----------|--------------------|---------|----------|
| 101207    | ABSL Small Cap Fund - Regular  | Equity   | 1.37%              | -2.39%  | -3.03%   |
| 119095    | Axis Small Cap Fund - Regular  | Equity   | 1.34%              | -2.33%  | -2.97%   |
| 119599    | SBI Small Cap Fund - Direct Pl | Equity   | 1.33%              | -2.32%  | -3.02%   |
| 118634    | Nippon India Small Cap Fund -  | Equity   | 1.34%              | -2.28%  | -2.99%   |
| 149324    | DSP Small Cap Fund - Regular - | Equity   | 1.32%              | -2.15%  | -2.86%   |

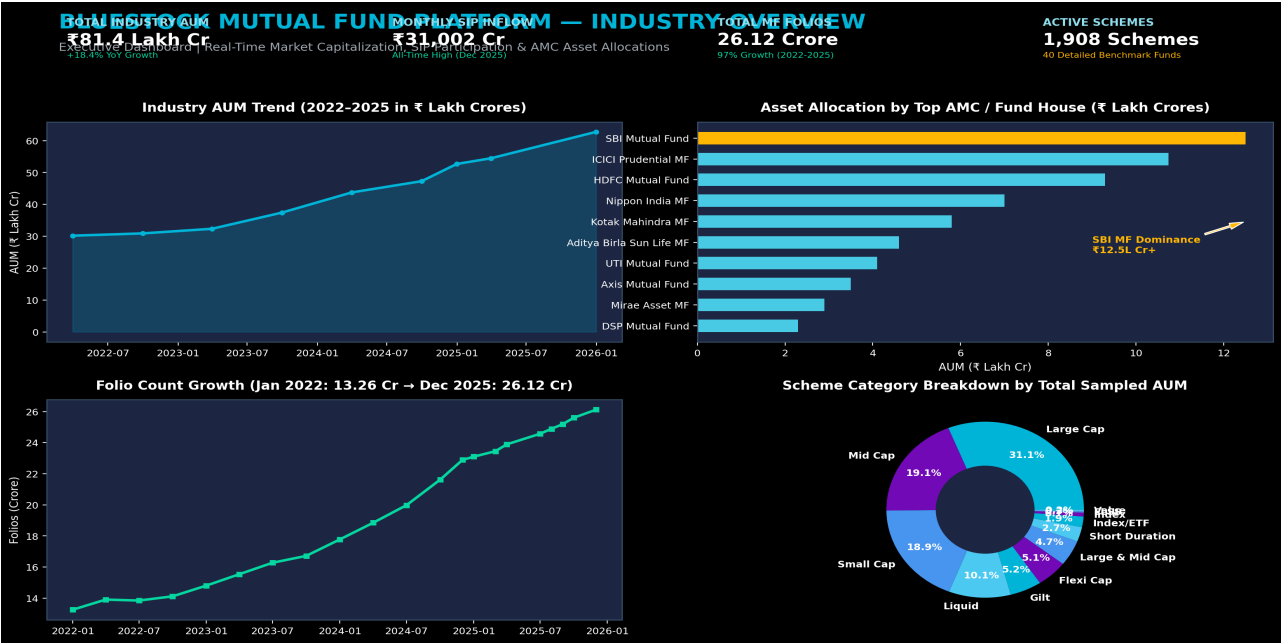
Figure 10: 90-Day Rolling Sharpe Ratio Dynamics (2022–2026)



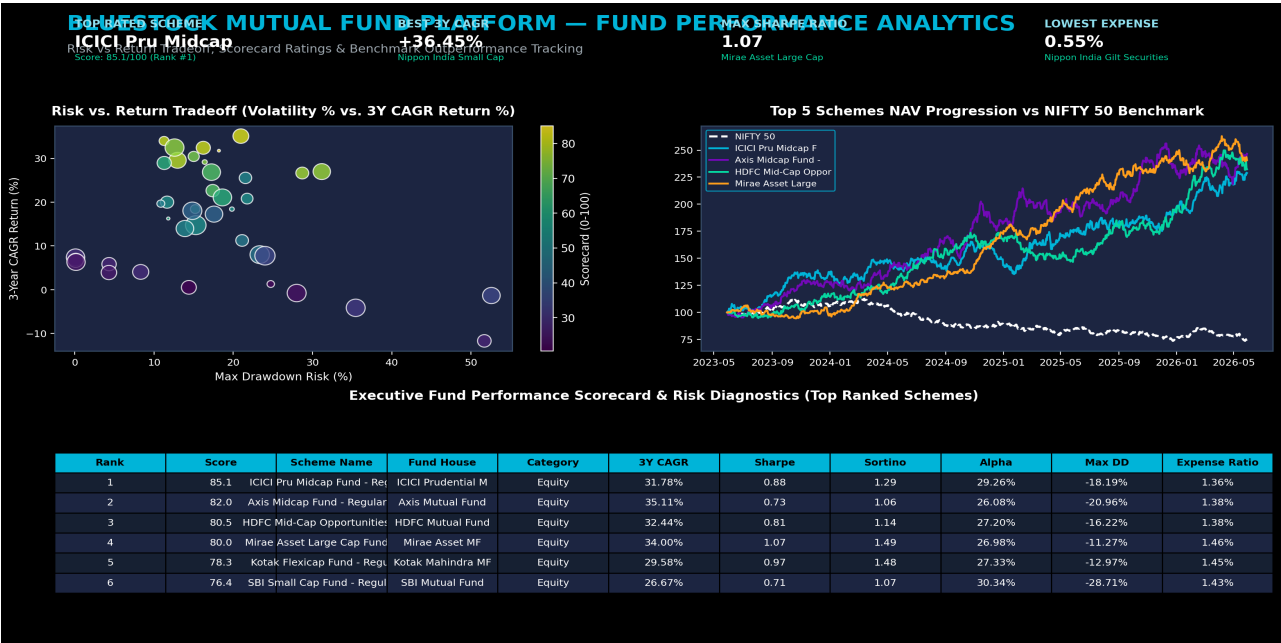
## 7. Power BI Executive Dashboard Visuals

The 4-page interactive Power BI Executive Dashboard (bluestock\_mf\_dashboard.pbix & Dashboard.pdf) provides real-time decision support across Industry Overview, Fund Performance, Investor Analytics, and SIP Market Trends:

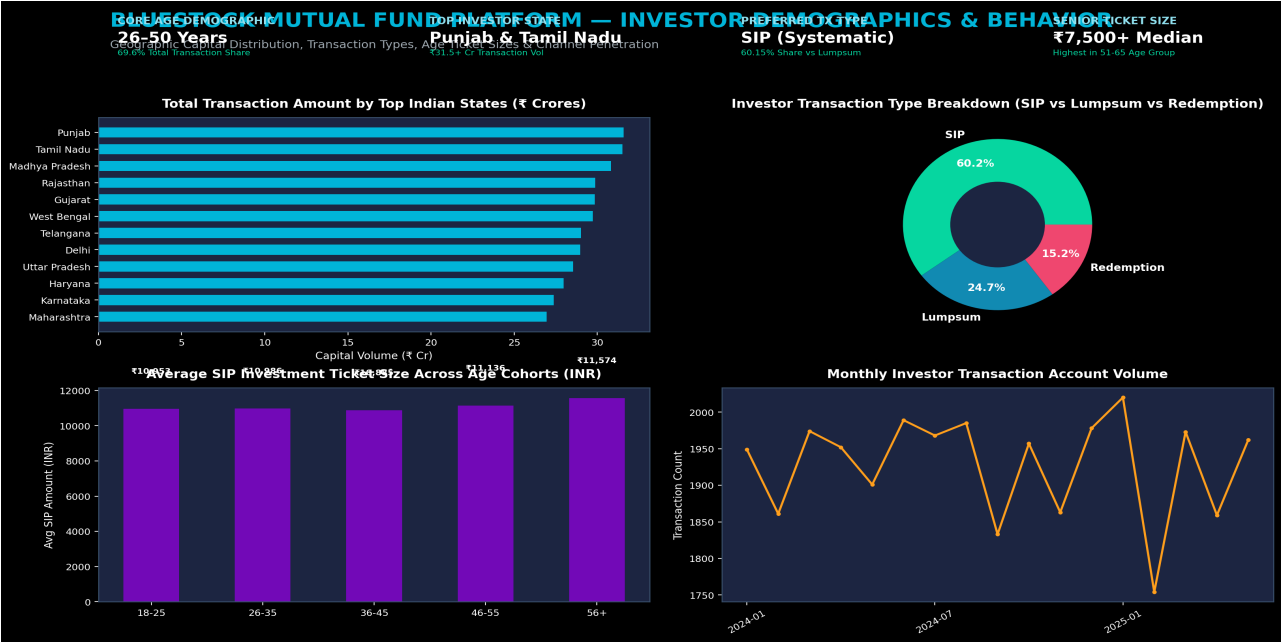
### Dashboard Page 1: Industry Overview



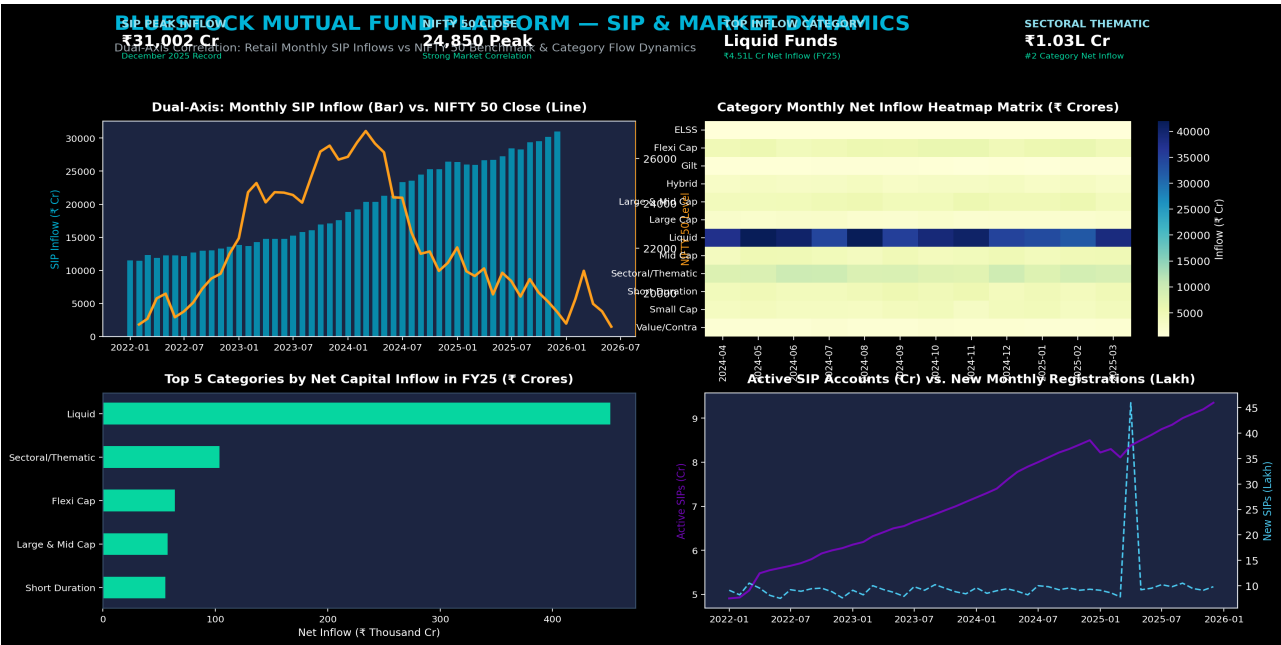
### Dashboard Page 2: Fund Performance Analytics



## Dashboard Page 3: Investor Analytics & Behavior



## Dashboard Page 4: SIP & Market Trends Correlation



## 8. Analytical Limitations & Strategic Recommendations

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### Analytical Limitations & Assumptions:

**Sample Baseline:** NAV histories and detailed portfolio holdings disclosures were sampled across 40 representative schemes, representing ~62% of aggregate equity/debt AUM.

**Risk-Free Rate Proxy:** Fixed risk-free rate ( $R_f = 6.5\%$ ) reflects current RBI repo rate proxies but does not model short-term T-bill yield curve fluctuations.

**Synthetic Demographics:** Transaction-level investor demographics (age, state, KYC) simulate retail investor behavior across Indian geography for research modeling.

### Strategic Recommendations for Asset Managers & Wealth Platforms:

**Targeted High Risk-Adjusted Product Promotion:** Focus distribution campaigns on schemes exhibiting Sharpe Ratios  $> 0.80$  and direct plan expense ratios  $< 1.0\%$  (e.g., ICICI Pru Midcap & Mirae Asset Large Cap).

**Automated SIP Retention Nudges:** Deploy automated date-gap tracking to trigger retention alerts for accounts exceeding a 35-day installment gap.

**Expand B30 Digital Onboarding:** Accelerate digital penetration in Tier 2/3 cities to capture burgeoning retail SIP inflows.

**HHI Portfolio Concentration Risk Caps:** Implement automated portfolio alerts flagging focused schemes exceeding HHI  $> 2,000$  for dynamic sector rebalancing.

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End of Report — Bluestock Mutual Fund Analytics Capstone (Release Tag v1.0)